

Engage AI and Child in Explanatory Dialogue on Commonsense Reasoning

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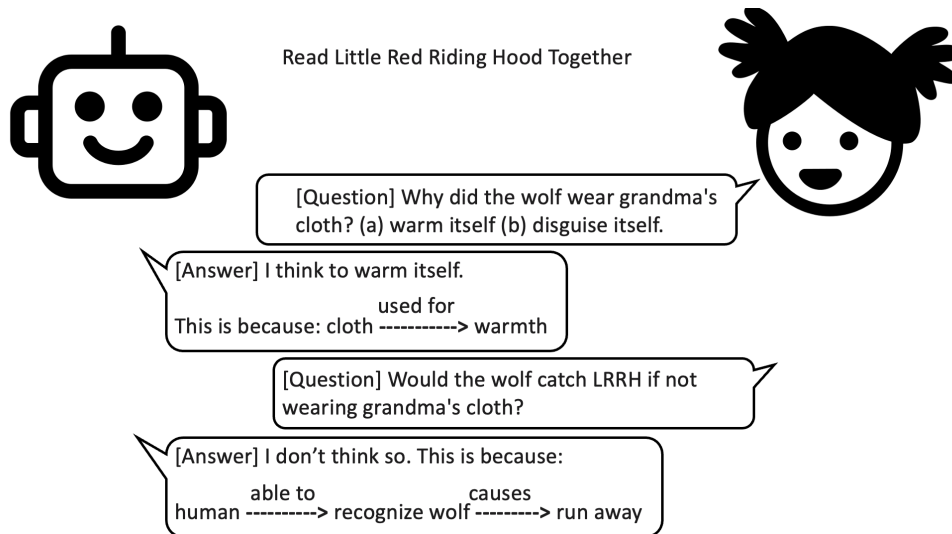


Figure 1: An idealized AI-child explanatory dialogue about the story of Little Red Riding Hood. The AI mistakenly inferred the answer to be *warmth* because it knew that *cloth can be used to keep warm*. The child tried to make the AI “realize” its mistake by inquiring about the consequence of the wolf not wearing the cloth. In the end, the child points out that *the wolf wore cloth to disguise itself*.

ABSTRACT

Human-level commonsense reasoning capability is vital for human-AI interaction, enabling AI to understand, anticipate, and respond to human’s thoughts, feelings, and behaviors. Despite the recent advancements in AI commonsense reasoning due to generative language models, a young child is often more rational than state-of-the-art AIs in terms of commonsense reasoning. The field of cognitive science, child development, and explainable AI have long recognized the importance of explanations for sharing knowledge and resolving contradictions. We, therefore, raise the question: *can*

AIs leverage the power of explanations to learn human-level commonsense reasoning? More specifically, can explanatory dialogue with children help AIs to develop commonsense reasoning capabilities? As a first step in this line of research, we aim to engage children in explanatory dialogue with AIs during story reading. We present our novel explanatory dialogue interface based on a state-of-the-art multi-step commonsense reasoning engine and discuss our upcoming pilot study.

CCS CONCEPTS

• **Human-centered computing** → **Collaborative interaction.**

KEYWORDS

Commonsense Reasoning, Explainable AI, Human-AI Collaboration

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1 INTRODUCTION

Since its early inception, the field of Artificial Intelligence (AI) has realized the need for intelligent programs to have common sense [39]. It is widely believed that common sense will play a key role in achieving human-level performance in several areas of AI [16]. From the human-AI interaction perspective, common sense—being able to act as a common ground for humans and AIs to establish shared mental states (e.g., thought, emotion, belief) [2]—will also likely play an important role in building trustworthy and teachable AIs [40]. Furthermore, multi-step commonsense reasoning could serve as easily interpretable explanations for laypersons to grasp the underlying causes of AI actions and valuable tools for diagnosing (or even correcting) unexpected or unacceptable AI behaviors.

A resurgence of interest in AI commonsense reasoning, primarily caused by the rapid advancement of generative language models, has led to significant progress in the area [12, 26, 48, 57] and related downstream tasks such as question answering [57] and dialogue response generation [86]. Nevertheless, a young child’s commonsense reasoning capability still often surpasses that of state-of-the-art AIs in terms of sophistication (i.e., the kinds of reasoning techniques employed), informativeness (i.e., the depth of the knowledge involved), and rationality (i.e., the overall logical correctness).

Social sciences such as cognitive science, child development and education have long recognized the importance of explanations for sharing knowledge and resolving contradictions [40]. Researchers in these fields propose various methods that use dialogic interaction and explanation to stimulate the development of causal reasoning [24, 50, 69, 78, 79]. Such explanations are often created using causal reasoning, which we view as an aspect of commonsense reasoning, and exchanged through dialogues. Informed by existing research on dialogical models of explanation in human-human conversation [74], recent research in explainable AI investigates the structure of explanatory dialog [40]. This research may serve as a general framework for creating conversational agents that engage human with explanatory dialog to improve AI commonsense reasoning.

We, therefore, raise the question *can AIs leverage the power of explanations to learn human-level commonsense reasoning?* More specifically, *can young children offer explanations for AI to develop commonsense reasoning capability?* As a first step in this line of research, we aim to engage children in explanatory dialogue with AIs during story reading. In this work, we present our novel explanatory dialogue interface based on a state-of-the-art multi-step commonsense reasoning engine designed for story comprehension tasks. We will describe the theoretical foundations and technical implementation for the components of the interface, and discuss our upcoming pilot study.

2 RELATED WORK

From the perspective of the explainer, the act of explaining involves two processes: *the cognitive process* identifies the causal relations of the situation to be explained, and *the social process* conveys the subset of the causal relations most relevant to the explainee [40]. In our work, the multi-step commonsense reasoner is responsible for the cognitive process, and the explanatory dialogue interface is responsible for the social process.

2.1 AI Common Sense

Large generative language models trained on massive corpora of natural language texts have drastically altered the field of natural language processing, achieving state-of-the-art results in numerous AI benchmarks, such as question-answering, summarization, and translation [47]. Many studies have investigated search-based multi-step commonsense reasoning where generative language models serve as generalizable commonsense knowledge resources [3, 27, 44, 77].

2.1.1 Generative Language Model. A generative language model utilizes search-based mechanisms to generate sequences of symbols following the probability distribution it has learned through training on massive text corpora (e.g., Figure 2). Such models demonstrate an understanding of not only the language’s grammatical structure but also world knowledge expressed in the training texts [28]. By carefully engineering the textual inputs for the model (i.e., prompt engineering) or additional training on task-specific examples (i.e., fine-tuning), generative language models have achieved state-of-the-art results in numerous tasks such as question answering and natural language generation [47].

2.1.2 Knowledge Triplet and Knowledge Model. Knowledge triplets are a kind of knowledge representation dominant in commonsense reasoning. Their simplicity allows them to be efficiently learned and queried. Knowledge triplets encode knowledge as head-relation-tail triplets (e.g., “Apples are fruits” can be encoded as (apple, is-a, fruit)) [61]. Bosselut et al. demonstrated that generative language models could serve as generalizable commonsense knowledge resources if fine-tuned using knowledge triplets [12]. Such triplet-generating models are often referred to as knowledge models [26], and they have been widely adopted in AI commonsense reasoning research.

2.1.3 Multi-Step Commonsense Reasoning. “By chaining together multiple commonsense knowledge triplets, one can form multi-step commonsense reasoning paths that express additional information beyond individual knowledge triplets [7]. ... Some generate reasoning paths using explicit traversals through commonsense knowledge graphs [27, 44]; others form reasoning paths by connecting commonsense knowledge triplets iteratively retrieved from triplet-generating language models [3, 77]” [76]. Paul and Frank demonstrated that by providing the generated multi-step commonsense reasoning paths as additional information to a classifier, which predicts the emotional state of characters of stories, the classifier gains in both interpretability and performance [44]. We adopt the multi-step commonsense reasoner proposed by Wang et al., which produces high-quality, contextualized, causal commonsense reasoning paths [76]. Wang et al. proposed *context extension* that uses a specialized knowledge model to incrementally augment a story by inserting sentences to highlight implicit commonsense information; by tracking the augmentations, context extension can generate multi-step commonsense reasoning paths [76].

2.2 Explanatory Dialogue

Although the theoretical analysis of explanatory dialogue by Walton [75] provides us with a general framework for building conversational agents to engage in explanatory dialogue, it needs more

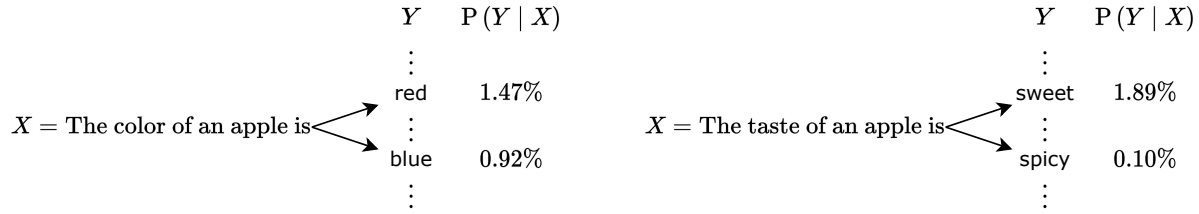


Figure 2: Next token probabilities calculated using HuggingFace’s implementation of gpt2 [82]. The model demonstrates its world knowledge by assigning a higher probability to the correct color and taste. Since there are tens of thousands of possible next tokens, each token receives a relatively small probability.

details, such as a specific task, for the dialogue to be purposeful. We, therefore, seek literature from other domains to equip us with those necessary details.

Commonsense reasoning, in the form of causal reasoning, has captured burgeoning attention in educational and psychological endeavors [6, 22, 53, 55]. Causal reasoning refers to human’s capacity of constructing cause-effect relations in the world around us, and it constantly evolves from early infancy [20, 22, 50]. Causal reasoning is crucial in child development because it allows children to make sense of the world, explain situations rationally, and function productively in academic settings [32, 56, 59, 62].

There are mainly three categories in causal reasoning [50]. Physical causality is a dimension towards children’s conception of interactions between physical objects. Situational causality is to address the connection between various story elements. The third category of causal reasoning is within the psychological domain, which links to human’s mental states. The cognitive competence to decode other people’s mental states such as their desires and beliefs is also widely known as theory of mind (ToM) [30, 31, 35]. We aim to help AIs achieve a better understanding of stories and protagonists’ emotions embedded in the story plots because such abilities ensure children to explain or predict other’s actions [30]. We, therefore, focus on the development of causal reasoning with a heavy emphasis towards its psychological domain.

2.2.1 Role of Dialogue for Reasoning. A rich literature has indicated that the development of causal reasoning in children results from dialogue with people around, such as their caregivers, peers and educational practitioners [50, 69, 79]. A series of studies show that parent-child conversation in the home setting can contribute to children’s development of reasoning towards others’ emotions and actions, because parents’ response and follow-ups during daily conversation could help children to enhance their thinking to the next level [25, 72, 79]. Other studies have repeatedly documented that narrative experiences, such as shared reading activities, create a learning space for children to exchange ideas and organize coherent causal structure with peers or teachers which develop their causal understanding of story elements [8, 50, 69].

During dialogic reading, a type of shared reading involving adult-child discussion on storybook related questions, interactive digital applications serve as a powerful tool to offer guidance for parents to generate questions, and utilize educational yet entertaining scaffolds to help children respond those questions [17, 29, 80]. The whole question-answer cycle around wordless picture books or

those with words can both trigger meaningful narrative production and advanced reasoning skills [29, 54].

In pedagogical interventions, social interaction within groups has been listed as one of the top efficient principles to establish students’ causal reasoning on history related academic tasks [45, 62]. Similarly, group discussion is considered as a powerful tool to generate collective discourse and high-level reasoning in science classrooms [41, 84]. Communication during such experiences could also foster children’s theory of mind capabilities, which help them interpret the mental state themes in storybooks [38, 58, 66, 85].

In addition to causality-oriented dialogue, visual (e.g., diagram, mindmap) and social representations (e.g., hand puppet, social robot) have been demonstrated efficient in demonstrating connections between logic information [13, 18, 70], conveying causal descriptions [1, 50], and stimulating peer discussion which leads to problem solving process and improvement of causal reasoning capabilities [33, 63, 64].

2.2.2 Role of Explanation during Dialogue. Why dialogic interaction promotes the development of causal reasoning? To answer this question, we can start with three layers within causal reasoning skills: causal inference is to draw interrelations between cause and effect [15, 21], prediction is to assume the effect or outcome of an event [11], and explanation is to describe the mechanisms resulting from the cause to its consequence [24, 33]. Explanation, in comparison with the other two dimensions of causal reasoning, has attracted increasing attention because it plays a fundamental role in the development of causal reasoning [79]. More specifically, it elicits more generalizable and abstract reasoning about the potential factor leading to its outcome [24, 34, 79].

Tracking back from early childhood, children are constantly interested in seeking and providing explanations regarding behaviors, minds, or emotions throughout their everyday conversation [14, 24, 78]. In children’s narratives, explanation often occurs along with the explicit markers of causality such as “why”, “how”, “because”, or their variations like “so”, “if”, “cause” [24]. Explanation allows children to generalize causal knowledge to novel situations efficiently [37, 73, 81]. When engaging in explanations with others, or simply to oneself, creates new learning opportunities for children to modify or further polish their causal knowledge and understanding during such iterative processes [60, 78].



Module: Loving Burgers

Story:
 Joe and his brother were craving burgers at midnight.
 Joe and his brother decided they were going to Mcdonald's.
 They went in Joe's car.
 They got burgers and fries with drinks.
 They no longer craved burgers.

Reasoning diagram:

```

    graph TD
      A[Joe and his brother go in Joe's car.] -- Results in --> B[Joe and his brother are at home.]
      C[Joe and his brother are getting burgers.] -- Causes --> D[Joe and his brother feel(s) happy.]
      B -- Results in --> D
  
```

Choices for the child:
☒ Repeat
☒ Explain it to me
☒ Move to next step

Figure 3: Our explanatory dialogue interface.

3 EXPLANATORY DIALOGUE INTERFACE

To investigate our research question: *can explanatory dialogue with children help AIs to develop commonsense reasoning capabilities?* we designed a Wizard of Oz (WoZ) system to enable explanatory dialogue between a child and a multi-step commonsense reasoner. The system consists of a web interface and a human operator. A child will vocally interact with the system using natural language, which the operator interprets.

The interaction is primarily carried out in the form of open-domain natural language conversation between a child and the interface's avatar (e.g., Figure 1). However, certain information, such as the AI's reasoning paths, may be conveyed via specialized visualization more efficiently, hence the need for the other interface components shown in Figure 3.

3.1 System Avatar

Existing literature in education has documented the effectiveness of harnessing robots to engage young children in the learning process, especially those with machine-like featureless appearances [10, 52, 68]. Robot with human-like appearance is less preferable to the child participants because it is less "friendly" and more "aggressive" compared to machine-like plain one [83]. In comparison with simply presenting tasks on screen-based devices, the involvement of interactive agents such as robot could provide a life-like illusion for the children, which would motivate their collaborative learning and establish social rapport [9].

To leverage children's affinity toward interactive agents, we built an avatar for the explanatory dialogue interface utilizing pre-drawn

faces. To avoid appearing unfriendly or aggressive, we designed the avatar to be more cartoon-like. During an interaction, the avatar can be in any of the following states:

- **Neutral state:** the base state of the avatar, activated when no interaction has occurred for a set period of time.
- **Emotion state:** one of the basic emotions defined by Ekman and Friesen (i.e., happy, sad, fear, anger, surprise, disgust), activated as a response to an interaction or at pre-defined times during story reading.
- **System state:** special states of the avatar, activated to communicate low-level system information to the user (e.g., to indicate that a computation or network request has taken longer than expected).

The avatar is implemented as a finite-state automaton that transition between states based on set timers, user interactions, and system exceptions. Although largely automated (i.e., the transition is done automatically based on the operator's inputs), the human operator can still manually place the avatar in any desired state.

3.2 Story Reader

The stories used during an interaction are pre-selected based on the emotional dimension, as we place a heavy emphasis on the psychological aspect of causal reasoning. Emotion can be defined as the "affective feelings caused by beliefs and desires" [51]. Using the framework proposed by Rashkin et al. [49], we annotate characters' emotions for stories of the ROCStories corpus [42], which our multi-step commonsense reasoner expects. As children develop an understanding of desire-based emotion (e.g., happy and sad)

earlier than belief-based emotion (e.g., fear and surprise) [5], we group stories based on the types of emotion (i.e., desire-based or belief-based) they exhibit. This grouping helps us ensure the age-appropriateness of the stories used during interactions. Finally, a domain expert filters the stories to ensure child-friendliness (i.e., age-appropriate language and no culturally unique element).

To help lower the children’s cognitive load, we ensure the story is constantly visible throughout the interaction. Furthermore, to help younger children during the interaction, pre-recorded audio reading of the story generated using commercial text-to-speech technology is made available, with playback controlled by the operator. In the rare cases that certain concepts of the story are unfamiliar to the child, the operator can enter a clarification that is then delivered to the child via text-to-speech.

3.3 Reasoning Diagram

The use of visual representation has been proven as an effective tool to store information in a structured knowledge system, thus reducing participants’ cognitive load and ensuring the understanding of the causality-related task [23, 36, 65]. Our interface adopts a simple block-based visualization for the reasoner’s reasoning paths. The operator is able to highlight parts of the reasoning path to direct the participant’s attention.

When multiple reasoning paths need to be visualized, the system will order the paths so that similar ones, as measured by textual and semantic similarity, are closer together. By placing similar reasoning paths in the proximity of each other, the participants can more easily compare and contrast between them, which Miller argues to be an important aspect of explanation [40]. The operator can manually reorder the reasoning paths if the automatic layout is undesirable; the operator can deliver the reasoning paths or any clarification via text-to-speech.

3.4 Formats of Child Feedback

To help improve AIs’ commonsense reasoning capability, we designed the interface to collect detailed feedback from the participants. We intentionally guide them through various types of questions, which are designed based on education and child development literature, to help children through the feedback process.

According to Pate [43], open-ended questions refer to those that are without restrictions on the quantity of correct answers. It is known as one type of high-level question to provoke deeper thoughts. Closed-ended questions, on the other hand, often have only one correct response. It is also useful because it offers an opportunity for participants to retrieve learned knowledge, especially those factual ones. Another classification approach during storytelling is to divide questions into literal or inferential ones. Literal questions are lower-level question types to identify or recall information in the story, while inferential questions are high-level and focus on information that is not in the story, such as things like students’ attitudes/feelings toward the story, connection to their experience/knowledge, prediction of the story, and etc.[71]. Although previous studies have documented more benefits to raising higher-level questions, such as open-ended or inferential questions for complex conversation and thought-provoking experiences, some have also suggested combining different question types together

for optimal learning outcomes [4, 43]. We have taken the latter approach when designing the interface.

4 LIMITATION AND FUTURE WORK

Although grounded in theories from cognitive science and child development, many of our design decisions are likely non-optimal, while the rest need to be field tested. There likely exist alternatives to our baseline block-based visualization that can better communicate AI’s reasoning and stimulate children’s thinking. Questions within the feedback form may require significant modifications and adaptations to better elicit diverse and insightful feedback from children. The involvement of a human operator prevents us from investigating genuine AI-child interaction. Future work could focus on integrating automatic speech recognition (e.g., whisper [46]) and dialogue response generation (e.g., LaMDA [67] and ChatGPT) into the system, thus eliminating or greatly reducing the need for a human operator.

To help validate our current interface’s design and inform our next iteration, we plan to conduct a pilot study to observe real-world interactions between our system and children from different age groups. Our intention for the pilot study is threefold:

- (1) To test our visualization technique and feedback elicitation mechanism: by observing how children respond to our interface, we can better understand its strength and weakness. This process will inform our next iteration of the system and help us devise a more systematic way of evaluating it.
- (2) To collect a corpus of AI-child interaction: a corpus consisting of real-world AI-child interactions would be beneficial. By analyzing the corpus, we could identify more efficient means of interaction and add them to the system (e.g., allow children to manipulate the reasoning paths via a touch display).
- (3) To collect a corpus of feedback: collecting a corpus of feedback would help us determine whether feedback can be used to improve AIs’ commonsense reasoning capability, if so, what kinds of feedback are desirable. The corpus could also allow the AI community to start investigating means of utilizing such feedback.

At its core, the proposed AI-child interaction aims to fulfill two objectives: the cognitive objective of eliciting high-quality commonsense knowledge and reasoning and the social objective of maintaining engagement with the participants. In terms of the cognitive objective, we intend to systematically transcribe the collected feedback into knowledge triplets and analyze their overlaps with existing commonsense knowledge resources, a common practice in commonsense knowledge graph research [26] that will enable us to evaluate the interface’s effectiveness at eliciting new commonsense knowledge. Regarding the social objective, human-AI collaboration in the form of open-domain natural language conversation is a new and exciting research area enabled by the development of large language models such as ChatGPT. However, although there has been much technical progress, more needs to be done to develop a formal evaluation framework for such interactions. Therefore, we wish that the pilot study would offer us some insights into the dynamics of such interaction and enable us to propose an evaluation framework for the engagingness of the conversation process.

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